HOMEWORK 11 FOR MATH 2860 (ELEMENTARY DIFFERENTIAL EQUATIONS)

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Due at 10:00 EST in class

1. For a 2×2 system:

$$\begin{cases} x_1' = 3x_1 + 4x_2 + e^{2t}, \\ x_2' = -x_1 + x_2 + \sin t, \end{cases}$$

   (a) Solve homogeneous system: $\mathbf{x}_h' = A\mathbf{x}_h, \quad A = \begin{bmatrix} 3 & 4 \\ -1 & 1 \end{bmatrix}$.

   (b) Find particular solution $\mathbf{x}_p(t)$ using undetermined coefficients or variation of parameters.

   (c) General solution: $\mathbf{x}(t) = \mathbf{x}_h(t) + \mathbf{x}_p(t)$.

2. Solve $\mathbf{x}' = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} \mathbf{x}$, $\mathbf{x}(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

3. Convert the IVP

$$y'' - 3y' + 2y = 0, \quad y(0) = 1, \ y'(0) = 0$$

   into a first-order system.

Show all your work.

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